Grant Wilkins

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Education

Stanford University Stanford, CA
Ph.D. in Electrical Engineering
- Advisors: Profs. Ram Rajagopal, Philip Levis
- Areas of Focus: Datacenters, Energy Systems, Cloud Computing

University of Cambridge, Churchill College Cambridge, UK
M.Phil in Advanced Computer Science July 2024
- Thesis: "Online Workload Allocation and Energy Optimization in Large Language Model Inference Systems"
- Advisors: Profs. Richard Mortier, Srinivasan Keshav

Clemson University Clemson, SC
Dual B.S in Computer Engineering and Mathematical Sciences May 2023
- Thesis: "Green HPC: Optimizing Software Stack Energy Efficiency of Large Data Systems"
- Distinctions: Norris Medal, Summa Cum Laude, General and Departmental Honors

Research Experience

Stanford University, Graduate Research Student Fall 2024–Present
Advisors: Richard Mortier, Srinivasan Keshav
- Developed an energy-aware online scheduling framework for LLM inference.

Microsoft Azure Research, Research Intern Summer 2025
Advisors: Fiodar Kazamiaka, Chaojie Zhang, Alok Khumbare, Ricardo Bianchini
- Explored and proposed designs for power and infrastructure layout for next-generation data centers.

Argonne National Lab, Graduate Research Assistant Summer 2024
Advisors: Sheng Di, Franck Cappello
- Quantified the energy-savings that lossy compression can introduce for exascale computing systems.

University of Cambridge, Graduate Research Student Fall 2023–Summer 2024
Advisors: Richard Mortier, Srinivasan Keshav
- Developed an energy-aware online scheduling framework for LLM inference.

Argonne National Laboratory, Graduate Research Assistant Summer 2023
Advisors: Sheng Di, Franck Cappello, Kibaek Kim
- Principal investigator for FedSZ: a lossy compression scheme to cut federated learning communication overhead.

Clemson University, Undergraduate Researcher Fall 2020 –Summer 2023
Advisor: Jon Calhoun
- Explored lossy compression and checkpointing for HPC towards cutting I/O energy and runtime overhead.

Industry Experience

Tesla, Inc., Software Engineering Intern — Summer 2021
Energy IoT Cloud Platforms Team — Palo Alto, CA

- Expanded functionality for the California Virtual Power Plant project on a fleet of Powerwalls.

Publications

[1] **G. Wilkins**, F. Kazhamiaka, and R. Rajagopal, "From Servers to Sites: Compositional Power Trace Generation of LLM Inference for Infrastructure Planning", 2026. arXiv: 2603.18383 `[cs.DC]`.

[2] S. Di, J. Liu, K. Zhao, X. Liang, R. Underwood, Z. Zhang, M. Shah, Y. Huang, J. Huang, X. Yu, C. Ren, H. Guo, **G. Wilkins**, D. Tao, J. Tian, S. Jin, Z. Jian, D. Wang, M. H. Rahman, B. Zhang, S. Song, J. Calhoun, G. Li, K. Yoshii, K. Alharthi, and F. Cappello, "A Survey on Error-Bounded Lossy Compression for Scientific Datasets", *ACM Comput. Surv.*, vol. 57, no. 11, Jun. 2025, ISSN: 0360-0300.

[3] **G. Wilkins**, S. Di, R. Calhoun Jon C. Underwood, and F. Cappello, "To Compress or Not To Compress: Energy Trade-Offs and Benefits of Lossy Compressed I/O", in *2025 IEEE 39th International Parallel and Distributed Processing Symposium (IPDPS)*, 2025.

[4] **G. Wilkins**, S. Di, J. C. Calhoun, Z. Li, K. Kim, R. Underwood, R. Mortier, and F. Cappello, "FedSZ: Leveraging Error-Bounded Lossy Compression for Federated Learning Communications", in *2024 IEEE 44th International Conference on Distributed Computing Systems (ICDCS)*, 2024, pp. 577–588.

[5] **G. Wilkins**, S. Keshav, and R. Mortier, "Hybrid Heterogeneous Clusters Can Lower the Energy Consumption of LLM Inference Workloads", in *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems*, ser. e-Energy '24, New York, NY, USA: Association for Computing Machinery, 2024, pp. 506–513, ISBN: 9798400704802.

[6] **G. Wilkins**, S. Keshav, and R. Mortier, "Offline Energy-Optimal LLM Serving: Workload-Based Energy Models for LLM Inference on Heterogeneous Systems", in *Proceedings of the 3rd Workshop on Sustainable Computer Systems*, ser. HotCarbon'24, New York, NY, USA: Association for Computing Machinery, 2024.

[7] **G. Wilkins** and J. C. Calhoun, "Modeling Power Consumption of Lossy Compressed I/O for Exascale HPC Systems", in *2022 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, Jun. 2022, pp. 1118–1126.

[8] **G. Wilkins**, M. J. Gossman, B. Nicolae, M. C. Smith, and J. C. Calhoun, "Analyzing the Energy Consumption of Synchronous and Asynchronous Checkpointing Strategies", in *2022 IEEE/ACM Third International Symposium on Checkpointing for Supercomputing (SuperCheck)*, Nov. 2022, pp. 1–9.

Teaching

- **Undergraduate Teaching Assistant** at Clemson University — Fall 2020 - Fall 2021
  Held office hours, graded, and led class sessions for over 50 students each semester in ENGR 1410.

Skills

- **Programming Languages:** C/C++, Python, Scala/Java, VHDL, FORTRAN
- **Tools/Software:** MPI, OpenMP, CUDA, PyTorch, MATLAB, LaTeX, Git

Honors and Awards

- **Churchill Scholarship** — 2023

- **Clemson University Norris Medal** 2023
- **National Scholars Program** 2019
- **Goldwater Scholar** 2022
- **Eagle Scout Award** 2016